

M2 IKSEM Session 1 – IT4-Blockchain Exam

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1 Questions on Blockchain Concepts (2 points for each question)

1. What are the advantages and disadvantages of POS?
2. How to minimize the trust requirement on an oracle?
3. In the Bitcoin network, does a miner have to revalidate all the blocks they receive? Why?
4. To make a payment in Bitcoin, do you need to have the exact amount? Explain why.
5. What are the features of blockchain systems that make them different from other data storage solutions?

2 Blockchain Patterns (10 pt)

2.1 Blockchain and Supply Chain

The horse meat lasagna scandal¹ caused consumer distrust towards mass-produced processed products. Indeed, many actors are involved in the supply chain: breeders, slaughterhouses, processors, industrialists, and distributors.

In case of fraud or possible contamination detected, for example, at a breeder, distributors must recall their products en masse, without being able to know with certainty which lots are involved, leading to significant food waste.

Similarly, some consumers need proof regarding the origin of the products, especially if they are endowed with specific quality labels (Organic, sustainable agriculture, high environmental value, etc.). They demand better product traceability, and a simple logo on the packaging is not enough: they need more tangible proof.

2.2 Questions

2.2.1 Advantages of a Blockchain Solution

As an innovative startup specializing in blockchain, you want to offer a solution to breeders, industrialists, and distributors to provide consumers with important information on the origin of the ingredients composing the final products and to optimize the recall of products impacted by a malfunction detected at any point in the supply chain.

During a farm visit, you meet Christelle DUCHENE (Master 2 in Nutritional Quality Management and Marketing of Food Products), who is the head of societal issues at INTERBEV². She is about to get into a tractor, dressed in overalls and boots: you only have 2 minutes to pitch your idea.

¹The company Spanghero is involved in the "Horsegate" scandal that broke out in January 2013. It is accused of fraudulently selling beef meat actually composed of horse meat to the Tavola factory in Capellen, a subsidiary of Comigel which supplies Findus. The company stated in a communiqué that it did not process the meat and sold it as is, blaming the Romanian slaughterhouses that originally supplied it.

²INTERBEV, the National Interprofessional Association of Livestock and Meat, is an interprofessional association and lobby grouping the main actors of the beef industry.

2.2.2 Solution Diagram

Your “tractor speech” was successful: you now have an appointment with Nicolas HENR, Internet and Multimedia Project Manager at INTERBEV (MIAGE Paris Saclay) to explain how to implement the solution you described to Christelle DUCHENE.

Using a component diagram, show how to deploy the solution using proven recipes (software patterns). Specify which blockchain technology could be used.

You can add a textual description of your answer.

2.2.3 List Relevant Patterns

Provide at least 3 of the most relevant patterns that can be used in the implementation of your solution. For each pattern, explain how and why it is used. Justify your answer using the properties of the patterns in relation to the use case.

1 - Oracle

Add an external trusted data source to trigger an event in the smart contract.

Oracle : Allows BC to use external data, can be implemented outside BC or use an injector within BC. Needs trust and validity, can use distributed oracle

2 - Reverse Oracle

Retrieve information from the blockchain to validate off-chain elements/conditions.

Reverse Oracle : Allows external system to use BC data by using SM to validate requests

3 - Legal & Smart Contract

Legal & Smart Contract aims to merge traditional legal contracts with smart contracts on a blockchain. Smart contracts automate transactions based on predefined conditions, ensuring speed and security without intermediaries. Meanwhile, Legal Contracts are real documents that establish the rights and obligations of the parties and are enforceable in courts outside the blockchain. This integration ensures that automated transactions comply with legal requirements, enhancing the efficiency of smart contracts with the legitimacy and enforceability of traditional contracts.

Example: A smart contract is used to manage apartment rent payments. On the blockchain, this smart contract sets specific rules. In parallel, a traditional legal contract details the rental terms and procedures in case of disputes. The smart contract references this legal contract, ensuring that automatic transactions adhere to legal terms, thereby guaranteeing legal compliance and transaction security.

4 - Encrypting On-Chain Data

This aims to preserve the confidentiality of sensitive data stored on a blockchain by encrypting it before recording. This process uses symmetric or asymmetric encryption to render the data unintelligible to anyone without the appropriate key. Keys should be managed off-chain and securely distributed only to authorized participants. This pattern addresses the transparency challenges inherent in blockchains by ensuring that, although data is technically publicly accessible, it remains confidential and inaccessible without decryption keys. This balances confidentiality requirements with the benefits of blockchain immutability and transparency.

5 - Tokenization

Use tokens to represent fungible assets.

- Native token: represents a monetary value or asset, allows tracking the transfer of assets (only physical).
- Define a data structure in a smart contract to represent assets.
- Tokenization process: involves locking an asset and representing it with a token.
- Disadvantages: legal processes for ownership (e.g., court decisions altering asset ownership without a blockchain transaction).

6 - Off-Chain Data Storage

Applications may want to use the blockchain to ensure the integrity of large amounts of data. Directly storing large data on the blockchain is not recommended due to limited block storage, limited scalability, and high data storage costs. Instead of storing raw data on the blockchain, a smaller value such as a hash value is recorded in a transaction, as a smart contract variable, or as a journal event. To retrieve the original data from the hash, one must have a copy of the raw data stored off-chain. By using the same hashing process with this

raw data, a new hash can be generated. If this new hash matches the hash stored on the blockchain, it guarantees data integrity and confirms that it has not been altered.

Advantages: integrity of values and reduced cost.

Disadvantages: security (data off-chain can be modified) / data loss.

7 - State Channel

- Solves the issue of micro-transactions (latency, transaction cost, limited scalability).
- Creates a communication channel between two participants: each micro-transaction is managed and added in a smart contract (off-chain).
- Settlement frequency is agreed upon by the participants and can be periodical, such as monthly.
- When the channel closes, the micro-transactions are aggregated into a single large transaction (final state), which is recorded on the blockchain.

8 - Multiple Authorization

Validates blockchain transactions by requiring multiple authorizations. A group of blockchain addresses is defined, but only some need to approve the transaction. This is useful for monetary transactions. It uses multi-signature (multi-sig), such as M-of-N, where M keys out of N are required, providing tolerance for lost private keys. This improves security and flexibility but can incur additional costs and requires defining all authorities in advance. Example: To authorize a transaction, three blockchain addresses (Alice, Bob, and Carol) are defined, but only two of them need to sign to validate the transaction. For instance, Alice and Bob sign the transaction, and it is approved without needing Carol's signature.

9 - Off-Chain Secret Enabled Dynamic Authorization

Uses an off-chain generated hash to dynamically bind transaction authority. Someone creates contracts with a lock that can only be opened with a key the other party possesses. At the time of locking, the keyholder is not defined. Essentially, there is no need to know who will have the key when creating the smart contract. The key can only be used once. Benefits : No need for specific private key, the key can be exchanged with any protocol Drawbacks : If the key is lost, the transaction will be locked forever

10 - X-Confirmation

Waits for a sufficient number of blocks as confirmation (6 => btc, 11 => eth) to ensure that a transaction added to the blockchain is immutable with high probability. Drawback=latency

11 - Contract Registry

A contract registry is a special contract that keeps a list of contract names and their addresses on the blockchain. It solves the problem of updating smart contracts on a blockchain by allowing the location of the latest version of a smart contract to be found by searching its name in a contract registry before invoking it.

12 - Data Contract

Context: Updating smart contracts can involve a high number of transactions if they contain a lot of data.

Objective: The goal of this pattern is to isolate DATA from the code in separate smart contracts.

Why use this pattern? When updating smart contracts using this pattern, only the smart contract code is updated, saving on gas and the number of transactions needed for the update.

13 - Embedded Permission

The proposed architecture aims to restrict access to smart contract functions to a set of authorized accounts. By default, contracts are accessible by all network participants, potentially compromising the network. This reduces potential vulnerabilities. By integrating authorization control mechanisms within the smart contract itself, it allows mapping participant addresses to a list to control interactions with smart contract functions (verification with the participant's public key). Note that the authorization control mechanism is implemented in the smart contract code.

14 - Factory Contract

Allows the creation of smart contract instances with customizable parameters. The factory contract itself remains unchanged and is highly secure as it is "on-chain." However, instantiating a smart contract from the factory contract and deploying the factory contract incurs costs.

Example: In a blockchain-based healthcare application, this pattern is used to create customized smart contracts for each patient, enabling efficient medical record management and better coordination among healthcare professionals.

15 - Incentive Execution

Incentive Execution is a blockchain pattern where a reward is provided to individuals who invoke certain smart contract functions, encouraging them to execute tasks that do not directly benefit them.

Pb: essential for the full functionality of the contract, usually incur costs without providing direct benefits to the users who trigger them

solution: Offer a reward to users who execute these functions, such as returning a portion of the transaction fee. This not only incentivizes users but also compensates them for the associated costs

ex: Ethereum Alarm Clock, this service schedules function calls for future blocks and provides incentives for users to perform these scheduled functions